PERFORMANCE CHARACTERISTICS TEST ON RECIPROCATING PUMP

AIM

To find out the efficiency of the reciprocating pump and plot the following graphs

1. Discharge v/s efficiency.
2. Discharge v/s head.
3. Discharge v/s Input power.

PRINCIPLE

Efficiency of the pump, η_p $= \dfrac{Power\ Ouput}{Power\ Input} \times 100\ \%$

Power Input, IP $= \dfrac{n\ 3600\ \eta_m\ \eta_T}{T\ E_m}$ kW

Where,

n = Number of revolution of energy meter (10 Rev)

T = Time taken for 10 revolution in sec.

E_m = Energy meter constant Rev/kW hour.

η_m = Efficiency of motor = 80 %.

η_T = Transmissions efficiency = 90 %

Output power, OP $= \dfrac{\rho\ g\ Q_{act}\ H}{1000}$ kW

Where,

ρ = Density of water = 1000 kg/m^3

g = acceleration due to gravity in m/s^2 (9.81 m/s^2)

Q_{act} = Actual discharge in m^3/sec. $= \dfrac{A\ R}{t}$ m^3/s

Where,

OBSERVATION AND TABULAR COLUMN.

	Deliv Pr	Speed	Deli Pr.	Suct Pres	Suct Pres	Total Head H	Time for 5 rev. 'T'	Time for 5 Cm t	Q th	Q act	OP	IP	Eff	Cd	% of slip
	kg/cm^2	rpm	m	mm. of Hg	m	m	s	s	m^3/s	m^3/s	kW	kW	%		
1															
2															
3															
4															
5															
6															

SAMPLE CALCULATION (Set No :......)

Input power, I.P $= \frac{n\ 3600}{T\ E_m} \eta_m \eta_T$ kW.

Number of imp of energy meter ,n = 5 Imp

Time for 'n' imp of energy meter, T = ... sec

Energy meter constant, E_m = ... Imp/kWh

Input power, IP = ... kW

Efficiency of motor, η_M = 80 %

Output power, OP $= \frac{\rho\ gQH}{1000}$ kW

Acceleration due to gravity, g $= 9.81 \frac{m}{Sec^2}$

Total Head, H = Suction Pressure + Delivery Pressure + Datum Head in m of water

Delivery Pressure head = ... $kg/cm2$ = __x10 m of water

Suction Pressure head $= \frac{......mm\ of\ Hg}{Density\ of\ water} = \frac{......}{1000\ X.1}$ 13.6 m of water

Datum Head = 350mm = 0.35m of water

Total Head, H = ... m of water

Discharge, Q $= \frac{AR}{t} \frac{m^3}{sec}$

Area of collecting tank, A = $0.25 \times 0.25\ m^2$

Height of water collected, R = 0.05 m

Time for collection, t = ... sec

Discharge Q $= \frac{AR}{t} \frac{m^3}{Sec} = \frac{0.25\ X\ 0.25\ X\ 0.05}{t} =$ .... $\frac{m^3}{s}$

Output power OP = ... kW

Overall Efficiency of Reciprocating pump $= \frac{Output}{Input} \times 100\ \%$

A	= Area of collecting in m^2	= $(0.25 \times 0.25)\ m^2$
R	= Rise of water in m (say 0.05m) in collecting tank.	
t	= Time in sec for the rise of 0.05 m of water in the tank.	
H	= Total head in m	

Total Head, H = Suction head(h_s) + Delivery head(h_d) + Datum head(h_z)

Coefficient of discharge, $C_d = \frac{Qact}{Qth}$ m^3/s

Theoretical discharge, $Q_{th} = \frac{L\,a\,N}{60}$ m^3/s

Where,

L= length of stroke in m (0.052 m)

a = Area of cross section of piston in $m^2 = \frac{\pi}{4}d^2$ (d =diameter of piston)

N= speed of pump in rpm.

Percentage of Slip = $(1 - C_d) \times 100$

PROCEDURE

1. Open the delivery valve.
2. Start the pump and allow the flow of water to stabilize.
3. Adjust the delivery valve to bring the delivery pressure to 0.5 kg/cm^2.
4. Note the reading of the suction pressure gauge.
5. Note the time't' taken to rise of water level 5 cm in the collecting tank.
6. Note the time 'T' taken for 10 revolution of energy meter disc.
7. Measure the speed of the crank.
8. Repeat this for delivery pressure readings 1 Kg/cm^2, 1.5, 2, 2.5 Kg/cm^2.
9. Switch off the pump.

GRAPHS

1. Discharge v/s efficiency.
2. Discharge v/s head.
3. Discharge v/s Input power.

RESULT

INFERENCE